Md. Mahin Rahman

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Education

Islamic University of Technology (IUT) — Gazipur, Bangladesh
B.Sc. in Electrical and Electronic Engineering (4th Year — CGPA: 3.64/4.00) — *2023 – 2027 (Expected)*

Govt. Edward College, Pabna | *Higher Secondary Certificate (Science) — GPA 5.00/5.00* — 2022

Pabna Zilla School, Pabna | *Secondary School Certificate (Science) — GPA 5.00/5.00* — 2020

Honors & Fellowships

CREST 2027 Research Fellow — Center for Research Excellence on Semiconductor Technology — 2027
- Selected as **one of 16 research fellows** across 10 projects and 8 universities nationally, in a programme run by Silicon River with Neural Semiconductor Ltd.; one of the undergraduate fellows, in the *Advanced Memory + 3D Integration* track.

Research Experience

Mismatch / Stack-Height Limits in 3D FeRAM — Gazipur, Bangladesh
CREST 2027 Research Fellowship — Advanced Memory & 3D Integration — *2026 – Present*
- Deriving mismatch-induced *stack-height limits* for current-mode non-destructive readout (NDRO) in three-dimensional ferroelectric memory arrays, as the selected project of a national semiconductor research fellowship.
- Analytical model linking ferroelectric capacitance mismatch, sense-amplifier transconductance and integration time to the maximum reliably readable stack height, backed by device- and circuit-level simulation.
- Three-researcher team with Mahir Faysal Sunny and Shihab Shahriar Hawlader; related work on floating-body effects and row-hammer disturbance in scaled 3D 1T-1C FeRAM arrays.

Undergraduate Research Assistant — Transportation Engineering — Gazipur, Bangladesh
Supervisor: Dr. Nazmus Sakib, PhD, PE — Dept. of Civil & Environmental Engineering, IUT — *2025 – Present*
- Design and build embedded instrumentation and data-acquisition systems for transportation research spanning vehicle braking dynamics, fleet tracking, roadside air quality and laboratory material testing.
- Instrumented a cycle rickshaw with encoder-based speed/distance measurement and time-of-flight brake-event detection, producing RTC-timestamped per-run datasets for braking-performance analysis.
- Developed laboratory readout tooling — multi-port load-cell acquisition and digital dial-indicator decoding — replacing manual reading with time-aligned CSV capture; also deployed a GPS-tagged particulate-matter logger and an RFID/QR rickshaw fleet system.

Optical IRS for Visible Light Communication (Li-Fi) — Gazipur, Bangladesh
Lead & Corresponding Author — Independent Simulation Research — *2026 – Present*
- Built an end-to-end Python channel-simulation framework for mirror-array optical intelligent reflecting surfaces (OIRS) in VLC, covering geometry, diffuse reflection, mobility, blockage and receiver noise.
- Identified two incompatible mirror reflection models in the OIRS-VLC literature and showed that omitting the law of reflection **overstates SNR by 6.3 dB**, with the error growing as the array scales.
- Derived a closed-form bisector orientation update that beats metaheuristic mirror-steering search by **five orders of magnitude** in compute, and showed control latency (13.0 dB) dominates blockage (1.1 dB) as a loss driver.
- Released a fully reproducible codebase — 12 publication figures, BER and power sweeps, and optimizer validation — at github.com/thisisdibbo/oirs-vlc-reflection-model; manuscript **under peer review** at *Optical and Quantum Electronics* (Springer).

Biodegradable / Transient Chips with On-Chip AI Integration — Gazipur, Bangladesh
Lead Author — Independent Simulation Research — *2026 – Present*
- Designing a simulation-only study of biodegradable transient semiconductor devices coupled with on-chip AI inference, targeting a Q1 journal.
- Device-to-circuit workflow spanning **Silvaco TCAD** process/device simulation and **Cadence Virtuoso** circuit design, with Python and MATLAB for extraction and analysis.

Publications & Manuscripts

M. M. Rahman, S. Tabassum, M. Mubasshirah, T. Hossain, and S. Nujabah, "Closed-Form Mirror Steering and the Cost of Control Latency in Optical IRS-Assisted Visible Light Communication," *Optical and Quantum Electronics* (Springer) — **under peer review**, 2026. *Corresponding author.*

Technical Skills

Programming Languages: C, C++, Python, MATLAB, Arduino C/C++, Kotlin (Jetpack Compose), LaTeX
TCAD, EDA & Simulation: Silvaco TCAD, Sentaurus TCAD, Cadence Virtuoso (Spectre netlist), Lumerical FDTD, MATLAB, Proteus, PSpice, COMSOL, PSIM, LINKPLANNER, CYME PSAF
CAD & PCB: SolidWorks, EasyEDA, EAGLE PCB, Tinkercad, 3D Printing
Hardware & Systems: ESP32, Arduino (Uno / Nano / Mega), Circuit Design, PCB Layout & Fabrication, Power Electronics, MOSFET & H-Bridge Motor Drivers, FPV Drone Build & Tuning
Sensors & Interfaces: GPS (NEO-6M / M8N), IMU (MPU6050), MAX30102, MLX90614, DHT22, PMS5003, VL53L0X ToF, MQ Gas, TDS / DS18B20, RFID, TFT & OLED, I^2C / SPI / UART / 1-Wire / ESP-NOW
Software & Data: PyTorch, NumPy / SciPy, PyQt6 / PySide6 Desktop Apps, Firebase Realtime Database, SQLite, SD / CSV Data Logging, Git & GitHub, PLC Ladder Logic (LS Electric XG5000), Betaflight, EdgeTX, GRBL
Languages: Bangla (Native), English (Professional Working Proficiency)

Academic & Engineering Projects

Smart Audio Recognition System | *Contributor — Python, PyQt5, DSP, DTW, SQLite* 2026
- Multi-modal music information retrieval engine that identifies original tracks, cover versions, instrumental renditions and hummed queries from live microphone input — 98% accuracy on clean audio, 67.5% on humming.
- Combines spectral-peak combinatorial hashing, PCEN-filtered chromagrams with dynamic time warping, and probabilistic YIN pitch tracking, behind a multithreaded PyQt5 interface over SQLite.

Smart Medical Box | *ESP32, ESP-NOW, TFT_eSPI, 3D Printing* 2025 – 2026
- Automated medicine cabinet: three motorised pill silos and two peristaltic syrup pumps, with a piezo disc under the tray so the box counts tablets it has *actually* delivered; MAX30102, MLX90614 and DHT22 vitals behind a 480×320 TFT, over a master–slave ESP-NOW link.

WaterSCADA — Water Quality Monitoring System | *ESP32, PyQt6, GPS, SCADA* 2025 – 2026
- GPS- and RTC-stamped logger measuring TDS/EC and DS18B20 water temperature to microSD CSV, with a PyQt6 SCADA dashboard — animated gauges, dark GPS track map, rolling trend charts — packaged as a single Windows executable.

Portable Air Quality Monitoring System | *ESP32, PMS5003, GPS, RTC, Firebase* 2025
- Portable particulate-matter logger: PMS5003 PM1.0/2.5/10 readings tagged with NEO-6M GPS position and DS3231 RTC time, logged to SD card in CSV; an earlier revision streamed live readings to a Firebase-backed web dashboard.

RFID Rickshaw Management System | *Python, Kotlin, Firebase, RFID* 2025
- Master/slave RFID pairing assigns a puller to a rickshaw in one transaction, across three linked clients — a Python desktop app, a Kotlin owner mobile app and a public QR lookup page — synchronised through Firebase Realtime Database.

Custom IBT-2 (BTS7960) H-Bridge Motor Driver PCB | *PCB Design, Power Electronics* 2025
- Designed, fabricated and bench-tested a custom high-current H-bridge driver board (output voltage equal to input under load), with an on-board HAT circuit — power LED, two signal LEDs and a JST header — for direct Arduino shield connection.

BreakProject — Braking Distance & Motion Logger | *ESP32, VL53L0X, Rotary Encoder* 2026
- Measures wheel speed, acceleration and travelled distance from a rotary encoder and detects brake events with a VL53L0X time-of-flight sensor, writing RTC-timestamped CSV logs per session; field-tested on a cycle rickshaw.

Additional Open-Source Work | *github.com/thisisdibbo* 2025 – 2026
- **Multi Serial Monitor** — PySide6 dashboard reading several QLB9 industrial load-cell indicators over simultaneous COM ports. **DialIndicatorReader** — Arduino firmware decoding the clock/data protocol of low-cost digital calipers. **Portable Display** — 3D-printed enclosure for a DIY portable monitor, released with STL files.

Digital Snake & Ladder Game | *Digital Logic Design, 74xx ICs, Proteus, PCB Fabrication* 2024 – 2025
- Decomposed a two-player board game into eight hardware modules — dice, player selector, player/snake/ladder position decoders, memory, winner detection, reset and board display — built from 74xx combinational and sequential logic in Proteus, with complete Gerber / CAD-CAM fabrication output.

FPV Drone Builds — Digital & Analog Video Systems | *ELRS, Betaflight, EdgeTX, RF* 2025 – 2026
- Built and tuned a quadcopter around a digital HD FPV camera and VTX link with an ExpressLRS control link; configured PID tuning, rates, failsafe and OSD, with EdgeTX setup on a RadioMaster TX16S.
- Built a second airframe on a 5.8 GHz analog camera/VTX chain — power distribution, ESC and motor wiring, antenna selection and full soldering assembly — then compared link latency and robustness in flight testing.

Laser Engraver with Custom Power Control Panel | *Power Electronics, GRBL, 3D Printing* 2025 – 2026
- Designed and fabricated an in-house power control panel board for a diode laser engraver — mains switching, regulated driver supply and protection circuitry — and 3D-printed the machine casing; driven through GRBL firmware with LaserGRBL toolpath control.

Industrial Training

Industrial Attachment Program Islamic University of Technology
Industrial Trainee *Sep. 2026 – Oct. 2026*
- **Ghorashal Power Station:** thermal power generation, generator synchronization, step-up transformers and high-voltage grid integration. **Bangladesh Atomic Energy Commission:** 5-day module on reactor physics, radiation safety protocols and advanced instrumentation.
- **Industrial Visits & Seminars:** cellular network architecture at Robi Axiata, automated PCB fabrication and floating solar PV at Walton Hi-Tech, and a VLSI design seminar by Neural Semiconductor Ltd.

Achievements & Awards

International Mars Rover Competitions | *Electrical Sub-Team Member, Project Altair* 2025 – 2026
- **University Rover Challenge (URC):** 2025 Finalist — placed **18th of 36** international teams in the world's premier Mars rover competition; 2026 On-Site Finalist.
- **International Rover Challenge (IRC):** on-site finalist in both the 2025 and 2026 editions. **Anatolian Rover Challenge (ARC):** 2026 on-site finalist, in autonomy, science and equipment-servicing missions.

Other Competitions | *IUT AUS Robo Race; PhysicsBrawl Online (Physics Olympiad)* 2023 – 2026

Leadership & Activities

IEEE Robotics and Automation Society, IUT Student Branch Chapter | *Joint Secretary* 2026 – Present
- **Instructor, Arduino Workshop (2026):** conducted a hands-on session for **50+ participants**, covering microcontroller fundamentals, sensor interfacing and embedded programming.
- Coordinate chapter operations, technical workshops and member activities across the robotics and automation community at IUT.

IUT Robotics Society | *Project Coordinator* 2023 – Present
- Manage technical workflows, competition preparation and hardware logistics across the society's research and competition teams.

Project Altair, IUT | *Senior Electrical Sub-Team Member* 2024 – Present
- Lead power distribution and protection design, embedded control and telemetry integration, PCB design and hardware testing for the team's competition Mars rovers.